

Since we assume T_j is $\mathcal{I}$ -model for $j = 1, \dots, m$, Lemma 2.13 induces

$$\begin{aligned} \int_X \langle \wedge_{j=1}^m T_{j,k}^D \rangle \wedge \omega^{n-m} &= \int_X \langle \wedge_{j=1}^m (\theta_j + \epsilon_k \omega)_{u_{j,k}^D} \rangle \wedge \omega^{n-m} \\ &\searrow \int_X \langle \wedge_{j=1}^m (\theta_j)_{P[u_j]_{\mathcal{I}}} \rangle \wedge \omega^{n-m} = \int_X \langle \wedge_{j=1}^m T_j \rangle \wedge \omega^{n-m}, \quad k \longrightarrow \infty. \end{aligned} \quad (3.10)$$

Combining (3.8), (3.9), (3.10) and let $k \longrightarrow \infty$. Then, we get (3.7)

$$\int_X \langle \wedge_{j=1}^m T'_j \rangle \wedge \omega^{n-m} \leq \int_X \langle \wedge_{j=1}^m T_j \rangle \wedge \omega^{n-m} - (\lambda'_m - \lambda_m) \prod_{j=1}^{m-1} \left(\frac{\delta_j \epsilon}{2c} \right)^{m-1} \int_{\widehat{X}} [\widehat{V}] \wedge (\sigma^* \omega)^{n-m} \wedge \widehat{\omega}^{m-1}$$

for the general case. Finally, since we assume that $T'_1, \dots, T'_m$ is of relative full mass intersection with respect to $T_1, \dots, T_m$, the inequality above does not hold and this make a contradiction. Therefore, $\lambda'_j - \lambda_j = \nu(T'_j, V) - \nu(T_j, V) = 0$ for some $j = 1, \dots, m$.

Step 3. We get rid of the assumption that V is smooth. By using Hironaka's desingularization method, we get $\sigma': X' \longrightarrow X$, which is a composition of finite many blow-ups along smooth centers, such that $V' = \sigma'^{-1}(V)$ is smooth.

Let $R_j = (\sigma')^* T_j$ and $R'_j = (\sigma')^* T'_j$. In general, $R'_1, \dots, R'_m$ do not necessarily have full mass intersection with respect to $R_1, \dots, R_m$. However, we have

$$\begin{aligned} \int_{X'} \langle \wedge_{j=1}^m R'_j \rangle \wedge ((\sigma')^* \omega)^{n-m} &= \int_X \langle \wedge_{j=1}^m T'_j \rangle \wedge \omega^{n-m} \\ &= \int_X \langle \wedge_{j=1}^m T_j \rangle \wedge \omega^{n-m} \\ &= \int_{X'} \langle \wedge_{j=1}^m R_j \rangle \wedge ((\sigma')^* \omega)^{n-m}. \end{aligned}$$

Since $[V] \wedge \omega^{n-m} \neq 0$, it follows that $[V'] \wedge (\sigma'^*(\omega))^{n-m} \neq 0$. Hence, we can apply **step 1.** and **step 2.** in this setting, and get

$$\nu(T'_j, V) = \nu(R'_j, V') = \nu(R_j, V') = \nu(T_j, V)$$

for some j . This completes the proof of Theorem 1.1.

3.2 Proof of Theorem 1.2

Let $\mathcal{B}$ be a closed cone in the cone of big classes, and let $V = \{x_0\}$ be a point in X . First, note that to prove inequality (1.1), it suffices to consider $\{\theta_j\} \in \mathcal{S} \cap \mathcal{B}$ for $j = 1, \dots, m$, where $\mathcal{S}$ is the unit sphere in $H^{1,1}(X, \mathbb{R})$. Recall that in (3.7), we choose

$$\delta_j = (c \|\{\theta_j\}\| + \frac{\epsilon}{2c_V})^{-1} \frac{\lambda'_j - \lambda_j}{2},$$

for $j = 1, \dots, m$. Here, ϵ is derived from the Kähler currents $P_j \in \{\theta_j\}$ such that $P_j \geq \epsilon \omega$. Since $\mathcal{S} \cap \mathcal{B}$ is compact, we can choose ϵ to be independent of $\{\theta_j\} \in \mathcal{S} \cap \mathcal{B}$. Also, by